QP Code 25901340

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Core Course (DSC) - MG4DSCICH200 - INSTRUMENTATION TECHNIQUES IN CHEMISTRY
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

(Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap))

Part A
Multiple choice questions Answer all questions. Each question carries 1 mark.

1.The time constant of an instrument indicates… [A] / [CO1]
b). Speed of response
2.Float type level indicators operate based on [U] / [CO1]
b). Buoyancy
3.Which of the following is an example of chemical noise? [U] / [CO1]
c). Reagent impurities
4.Which type of spectroscopy involves transitions between electronic energy levels? [K] / [CO2,CO3]
c). UV–Visible spectroscopy
5.Which device disperses light into its component wavelengths and allows selection of a narrow wavelength band? [K] / [CO2]
b). Monochromator
6.The signal produced by a detector is usually proportional to: [U] / [CO2]
c). Sample concentration
7.What is the unit of specific rotation in polarimetry? [U] / [CO4]
d). Degrees/(g/mL)
8.In a polarimeter, what is the primary function of a polarizer? [A] / [CO4]
d)To select and transmit light of a specific polarization.
9.While measuring the refractive index of a volatile liquid, the reading fluctuates continuously, which instrument component should be checked first? [A] / [CO4]
b). Water circulation system
10.Beer–Lambert law is applied in UV–Visible spectroscopy to determine: [U] / [CO3]
b). Concentration of solution

Part B
Short Answer Questions Answer any 4 questions
Each question carries 3 marks

11.Explain how automatic instruments reduce human error. [A] / [CO1]
Automatic instruments reduce human error by minimizing direct human involvement in measurement and data handling processes. They eliminate observational errors such as parallax by providing precise digital readings, and they avoid mistakes in recording and calculation through built-in data logging and processing systems. These instruments ensure consistent and repeatable results by operating under fixed conditions, independent of human fatigue or skill level. They also respond quickly to changes and enable continuous monitoring, reducing the chance of missed observations. Moreover, automatic instruments remove personal bias by relying on objective measurements, thereby improving the overall accuracy and reliability of results. (Any relevant three points – 3 marks)

12.What is time constant? Explain its importance in first-order instruments.[A] / [CO1]
The time constant is the time required for a first-order instrument to reach about 63.2% of its final steady-state value after a sudden change in input. It is a measure of the speed of response of the instrument. The importance of the time constant lies in its ability to indicate how quickly an

instrument can respond to changes in the measured quantity. A smaller time constant means a faster response and better ability to track rapid changes, while a larger time constant indicates a slower response and possible lag in measurement. First-order instruments are measuring devices whose response to a change in input follows a first-order differential equation, meaning the rate of change of the output is proportional to the difference between the current output and the final steady value. When subjected to a sudden change in input, the output does not respond instantaneously but approaches the final value gradually in an exponential manner. These instruments are characterized by a single time constant, which determines the speed of response. (Any relevant three points – 3 marks)

13.Calculate the energy of radiation having wavelength 400 nm. (h = 6.626 × 10 ³ J·s, c = 3 × 10 m/s). [A] / [CO2,CO3]

Energy, $E = \frac{hc}{\lambda}$

$$E = \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{400 \times 10^{-9}}$$

$$E = \frac{19.878 \times 10^{-26}}{4 \times 10^{-7}}$$

$$E = 4.97 \times 10^{-19}$$

(Equation – 1 mark, Correct substitution – 1 mark, Final answer-1mark)

14.How is IR spectroscopy used to identify functional groups? [U] / [CO3]
Infrared (IR) spectroscopy is used to identify functional groups based on the absorption of infrared radiation, which causes vibrations of chemical bonds in a molecule. Different functional groups absorb IR radiation at characteristic wavenumbers due to their specific bond strengths and atomic masses. The IR spectrum displays peaks corresponding to these absorptions, and by comparing them with standard values, functional groups such as –OH, –NH, C=O, and C–H can be identified. In addition to the functional group region (4000–1500 cm ¹), the fingerprint region (1500–400 cm ¹) is also important, as it contains a complex pattern of absorptions unique to each compound. This region helps in confirming the identity of a substance by matching it with known spectra, making IR spectroscopy a reliable method for both identification and structural analysis. (Any relevant three points – 3 marks)

15.State Snell's law and write its mathematical expression. [U] / [CO4]
Snell's law states that when a ray of light passes from one medium to another, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant for a given pair of media.
The mathematical expression is $n_1 Sini = n_2 Sinr$ where i is the angle of incidence, r is the angle of refraction, and n1 and n2 are the refractive indices of the two media.
(Statement – 1.5 marks, Mathematical expression with terms – 1.5 marks)

16.What is the function of the reference material holder? [U] / [CO4]
The reference material holder is used to hold a standard or reference sample (such as a blank or solvent) against which the sample is compared during measurement. Its function is to account for absorbance due to the solvent, container, or instrument, thereby eliminating background interference and ensuring accurate results. In a double beam spectrometer, the reference material holder plays an essential role by continuously comparing the reference and sample beams simultaneously. This allows real-time correction of fluctuations in source intensity and instrument conditions, leading to more precise, stable, and reliable measurements.
(Any relevant three points – 3 marks)

Part C

Short Essay Questions Answer any 4 questions
Each question carries 7 marks

17.Discuss the significance of S/N ratio in analytical techniques such as spectroscopy and chromatography. [U] / [CO1]

Signal refers to the measurable response produced by the analyte of interest, while noise refers to unwanted random fluctuations or disturbances that affect the measurement. The signal-to-noise (S/N) ratio is a measure of the strength of the analytical signal relative to the background noise, and it plays a crucial role in techniques such as spectroscopy and chromatography. A high S/N ratio indicates that the signal is clearly distinguishable from noise, leading to more accurate and reliable results, while a low S/N ratio makes detection and quantification difficult. In spectroscopy, a higher S/N ratio improves the clarity of spectral peaks and enables detection of low concentrations of substances. In chromatography, it enhances peak identification, resolution, and accurate integration of peak areas. The S/N ratio also determines important analytical parameters such as limit of detection (LOD) and limit of quantification (LOQ). Thus, improving the S/N ratio is essential for achieving high sensitivity, precision, and overall analytical performance.

(Definition and equation – 4 marks, Significance – 3 marks)

18.What is a flicker noise? What are its characteristics? What are the methods to reduce it? [U] / [CO1]

Flicker noise, also known as 1/f noise, is a type of electronic noise that occurs in many electrical and electronic devices, especially at low frequencies. It arises due to fluctuations in the flow of charge carriers, often caused by defects, impurities, or irregularities in materials and interfaces. The name 1/f noise comes from the fact that its power spectral density is inversely proportional to frequency, meaning it becomes more significant at lower frequencies.

The main characteristic of flicker noise is that its magnitude increases as the frequency decreases, dominating over other types of noise such as thermal noise at low frequencies. It is present in devices like transistors, resistors, and semiconductor components. Flicker noise is generally independent of temperature to some extent but strongly depends on the quality of the material and manufacturing process. It is more prominent in semiconductor devices due to carrier trapping and release mechanisms at defect sites.

To reduce flicker noise, high-quality materials with fewer defects and impurities should be used during fabrication. Proper device design and improved manufacturing techniques can minimize imperfections that cause noise. Increasing the operating frequency can help, as flicker noise is less significant at higher frequencies. Using filtering techniques to eliminate low-frequency components and employing signal modulation methods such as chopper stabilization or auto-zeroing can effectively reduce its impact. Additionally, maintaining stable operating conditions and using proper shielding and grounding can further help in minimizing flicker noise in practical circuits.(Definition with characteristics /equation – 4 marks, Reduction methods -3 marks)

19.Write short notes on: (a) Nernst Glower (b) Globar Source. [U] / [CO2]

(a) Nernst glower is a commonly used infrared radiation source in infrared spectroscopy. It consists of a small cylindrical rod made of a mixture of rare earth metal oxides such as zirconium oxide, yttrium oxide, and thorium oxide. At room temperature, these oxides are non-conductors, so the glower cannot conduct electricity initially. Therefore, it requires preheating using an external heater to raise its temperature to a point where it becomes electrically conductive. Once it becomes conductive, an electric current is passed through the rod, heating it to a high temperature of about 1000–2000 °C. At this elevated temperature, it emits continuous infrared radiation over a wide wavelength range, typically from about 2 to 15 μm. The emitted radiation is fairly intense and stable, making it suitable for spectroscopic measurements. The Nernst glower has a relatively long operational life and provides higher intensity radiation compared to some other sources in certain regions. However, it is fragile, requires a preheating arrangement, and its operation is somewhat more complex due to the need for temperature control and initial heating. (3.5 marks)

(b) Globar source is another important infrared radiation source used in infrared spectroscopy. It is made of silicon carbide in the form of a rod or cylindrical element. Unlike the Nernst glower, silicon carbide is electrically conductive at room temperature, so it does not require any preheating to start

operation. When an electric current is passed through the globar, it is heated to temperatures around 1200–1500 °C, resulting in the emission of continuous infrared radiation. The radiation produced covers a broad wavelength range, typically from about 2 to 20 μm. The globar source is mechanically strong, robust, and easy to handle compared to the Nernst glower. It provides stable and reproducible radiation and is widely used in modern infrared spectrometers because of its simplicity and reliability. Although its intensity may be slightly lower than that of the Nernst glower at shorter wavelengths, it has the advantage of uniform emission and longer operational stability. Additionally, it does not require complex starting procedures, making it more convenient for routine laboratory use. (3.5 marks).

20.Derive and explain the Beer–Lambert law.[U] / [CO3]
Beer–Lambert law states that when a beam of monochromatic light passes through a solution, the amount of light absorbed is directly proportional to the concentration of the absorbing species and the path length of the medium. Mathematically, it is expressed as $A = \varepsilon cl$, where A is the absorbance, ε is the molar absorptivity, c is the concentration of the solution, and l is the path length. As the concentration of the solution increases, more light is absorbed, leading to higher absorbance. Similarly, an increase in the path length also results in greater absorption of light. The absorbance is also related to the intensity of incident and transmitted light by the relation $A = \log(I / I)$. This law is valid under conditions such as the use of monochromatic light, dilute solutions, and absence of scattering or chemical changes in the solution.
(Statement – 3 marks, Derivation -4 marks)

21.Define Thermal Gravimetric Analysis (TGA). Explain its principle, instrumentation and working in detail. [E] / [CO4]
Thermal Gravimetric Analysis (TGA) is a thermal analysis technique in which the mass of a substance is measured as a function of temperature or time under a controlled atmosphere.
The principle of TGA is based on the fact that many materials undergo changes in mass when heated due to processes such as decomposition, oxidation, reduction, or loss of volatile components. By continuously recording the weight change as the temperature increases, information about thermal stability and composition of the sample can be obtained.
The instrumentation of TGA mainly consists of a sensitive microbalance, a furnace, a temperature programmer, a sample holder, and a gas flow system. The microbalance continuously measures the weight of the sample with high precision. The furnace provides a controlled heating environment, and the temperature programmer regulates the rate of temperature increase. The sample holder, usually made of inert material like platinum, holds the sample. The gas flow system supplies inert or reactive gases to control the atmosphere during the analysis.
In the working of TGA, a small amount of sample is placed in the sample holder attached to the microbalance. The sample is then heated at a controlled rate in the furnace under a specified atmosphere such as nitrogen, air, or oxygen. As the temperature increases, any change in mass due to decomposition, evaporation, or chemical reactions is recorded continuously. The result is plotted as a thermogram, which shows weight change versus temperature or time. From this curve, important information such as moisture content, thermal stability, decomposition temperature, and composition of the material can be determined.
(Principle – 2 mark, Instrumentation -3 marks, Working – 2 marks).

22.Write a note on different sampling methods. [U] / [CO3]
Sampling is an important step in analytical techniques, as it ensures that the portion of material taken for analysis accurately represents the whole sample. Different sampling methods are used depending on the nature of the material and the purpose of analysis.

- Random sampling involves selecting portions of the sample at random, ensuring that every part has an equal chance of being chosen. It minimizes bias and is widely used when the sample is homogeneous.
- Systematic sampling is carried out by selecting samples at regular intervals, either in time or space. It is simple and useful when dealing with large bulk materials or continuous production processes.

- Stratified sampling is used when the material is heterogeneous. The sample is divided into different layers or strata, and samples are taken from each layer to ensure proper representation of all components.
- Grab sampling involves taking a single sample at a specific time or place. It is quick and convenient but may not represent the entire bulk accurately.
- Composite sampling is done by combining several individual samples taken from different locations or times. This method provides an average representation of the entire sample and is commonly used in environmental and industrial analysis.

  (Sampling definition and importance – 2.5 marks, Explanation of any three sampling methods – 4.5 marks)